Singapore Deep-Tech Alliance

SG Eco Loop 2.0

Nationwide Quest for the Next Generation of Sustainable Innovators

Kick-off

26 Aug 2026

Supported by

SG Eco Fund · intel · Lenovo

Part of

SG Youth Plan · NYC National Youth Council Singapore

Catalysed by

OSCAR · Temasek Foundation · SCAPE

In Support of

Go Green SG · Switch Singapore Week of Innovation & Technology by Enterprise Singapore · SG 60

SG Eco Loop

Equips youths with the capabilities, networks, and opportunities to use innovation to tackle some of the most pressing challenges facing our planet

An SDTA programme

Singapore Deep-Tech Alliance

Non-profit organisation bringing together people, ideas, and partnerships to advance innovation for a more inclusive and sustainable future.

Singapore Deep-Tech Alliance

HOUSEKEEPING

- Rename yourself to:
  - First Name (Team)
  - e.g. Jamie (Eco)
- Keep your cameras on and mute audio unless speaking
- Raise hands / ask questions in chat anytime
- Be respectful of one another
- Competition, but also learning environment

COMMUNICATIONS

WhatsApp

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- Email: Add sgecoloop@sdta.org.sg to safe sender list

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OBJECTIVE

Build prototype(s) to divert food waste & plastic waste from landfill

Physical

Digital

Community Initiative

Redesigned workflow / system

looks automated to the user but a team member is secretly doing the work behind the scenes (e.g. a "smart" sorting bin controlled manually

Wizard of Oz prototype

Incentive Model

Good representation of your proposed solution that you can use to test, get feedback and present

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HACKATHON CHALLENGE & PRIZES

Biggest Impact
S$1,000

- Most kg of plastic and/or food waste diverted from landfill
- Diverted: reduced, repurposed, recycled
- Need not be directly attributed to your prototype(s)

Best Innovation
S$1,000

Decided by mentors jury panel based on 5 criteria:

- **Problem understanding** - how clearly the team has grasped a real plastic or food waste problem and who it affects
- **Originality** - how fresh the approach is compared to existing solutions
- **Feasibility** - how realistic the prototype is to build, test, and actually use, given the team's resources
- **Potential impact** - how much waste could be diverted or behaviour changed if the idea scaled beyond the programme
- **Pitch and clarity** - how well the team communicates their problem, solution, results and scaling plan at showcase

DELIVERABLES

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Prototype

- Showcase at Temasek Shophouse on **2 Oct**
- Physical, digital, or both - solving a plastic / food waste problem
- Get creative with your display: poster board, video loop, photos, etc
- Show your journey and **win Best Innovation**!

Scale-up Plan

- Learnings, how you'd scale up / pivot your innovation if you could continue building this
- Take a longer term view to problem solving

Trash Diaries

- Log every kg diverted each week - weigh & photo
- Aim for 20 kg or more per team
- **Win Biggest Impact**!

People Log

- Log all individuals you engaged during your journey
- Share about SG Eco Loop, waste diversion journey, survey participants, test users, encourage recycling, etc

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Prototyping Budget

S$200
per prototype

To unlock the budget for a second prototype, they need to be conceptually different

Ask us anything, anytime!

White-listed Expenses (Click for Full List)

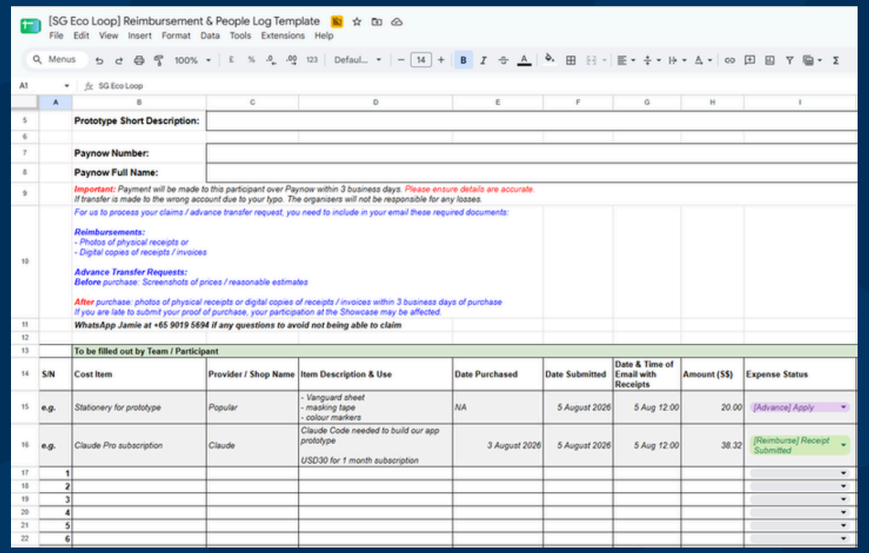
- Stationery, AI (Claude, Gemini, ChatGPT), Canva, 3D Design, 3D Printing

Non-white-listed Expenses

- Email us with short description, rationale, cost

Claims Process

- Update reimbursement form anytime and email us
- Add on to the same Google Sheet for multiple submissions
- Paynow within 1-3 business days
- Latest submission: 1 October



[SG Eco Loop] Reimbursement & People Log Template

File Edit View Insert Format Data Tools Extensions Help

A1 | SG Eco Loop

Prototype Short Description:

Paynow Number:

Paynow Full Name:

Important: *Payment will be made to this participant over Paynow within 3 business days. Please ensure details are accurate.*
If transfer is made to the wrong account due to your typo. The organisers will not be responsible for any losses.

For us to process your claims / advance transfer request, you need to include in your email these required documents:

Reimbursements:
- Photos of physical receipts or
- Digital copies of receipts / invoices

Advance Transfer Requests:
Before *purchase: Screenshots of prices / reasonable estimates*

After *purchase: photos of physical receipts or digital copies of receipts / invoices within 3 business days of purchase*
If you are late to submit your proof of purchase, your participation at the Showcase may be affected.

WhatsApp Jamie at +65 9019 5694 if any questions to avoid not being able to claim

To be filled out by Team / Participant

S/N	Cost Item	Provider / Shop Name	Item Description & Use	Date Purchased	Date Submitted	Date & Time of Email with Receipts	Amount (S$)	Expense Status
e.g.	*Stationery for prototype*	*Popular*	*- Vanguard sheet - masking tape - colour markers*	*NA*	*5 August 2026*	*5 Aug 12:00*	*20.00*	*[Advance] Apply*
e.g.	*Claude Pro subscription*	*Claude*	*Claude Code needed to build our app prototype USD30 for 1 month subscription*	*3 August 2026*	*5 August 2026*	*5 Aug 12:00*	*38.32*	*[Reimburse] Receipt Submitted*
1								
2								
3								
4								
5								
6								

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Kahoot!

How much do you know about Singapore’s waste challenges?

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So how can we tackle waste?

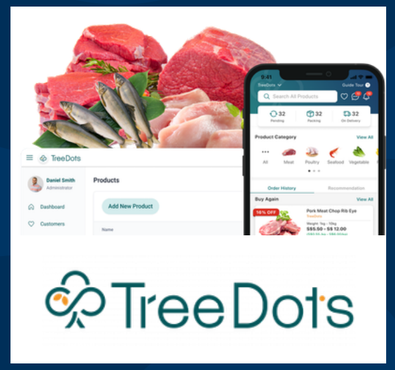
Reduce,
Reuse,
Recycle
+6 other ‘R’s

9 “R”s

1. Refuse

Say no to the material in the first place.

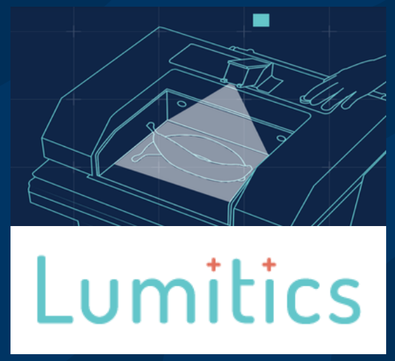
BYO Singapore - the Zero Waste SG campaign that gets shoppers and diners to bring their own bag, bottle or container and skip the disposable. Participating retailers give small rewards.



2. Rethink

Redesign the model so less is wasted.

TreeDots - a B2B marketplace that reroutes suppliers' unsold or "imperfect" stock to buyers, instead of letting it expire in the supply chain.



3. Reduce

Produce or use less in the first place.

Lumitics - their "Insight" device sits over a kitchen bin and uses AI and a camera to track what a buffet or hotel throws away, so chefs can cut over-preparation.
(Acquired by Winnow)

9 “R”s

4. Reuse

Use the same item again.

Muuse - a returnable cup and container rental system running in Singapore hawker centres and offices. Take your food or drink in a reusable, return it, and it gets washed and recirculated.

5. Repair

Fix it so it lasts longer.

Repair Kopitiam - a Sustainable Living Lab initiative where volunteer "repair coaches" help residents mend broken electronics, appliances and household items at neighbourhood sessions instead of throwing them out.

6. Refurbish

Restore an old product to working order.

Engineering Good - a charity that collects and refurbishes used laptops and computers, then passes them to low-income families and students, keeping working electronics out of the e-waste stream.

9 “R”s

7. Repurpose

Turn it into something with a new use.

The Interchange - Singapore's circular economy lab that turns everyday waste like rice sacks, drink cartons and offcuts into new handmade products through research, workshops and their own shop.

8. Recycle

Break it down and make it into something new.

Magorium - a Singapore company that recycles plastic waste into NEWBitumen, a road-paving material now used on real roads across the island, including at Tuas.

9. Recover

Get energy or material from what's left.

Tuas Nexus - Singapore's upcoming integrated waste and water facility, where incinerable waste and food waste sludge are converted into biogas and electricity, powering the plant itself.

muuse

Making Reuse Second Nature

We make it easy to borrow, return and reuse packaging — instead of throwing it away.

END-TO-END SERVICE

- Hardware: Smart Return Infrastructure, Reusable Packaging
- Washing & Logistics service
- Data platform for reporting, inventory + incentives/rewards.

WHERE WE WORK

- Large-scale events
- Semi-closed spaces (Workplace, Schools, Tourism)
- Cities

2M single-use items diverted from landfill
200 tonnes carbon avoided

Jonathan Tostevin
CEO, Muuse

Singapore Deep-Tech Alliance

Next:

Double Diamond

HOUSEKEEPING

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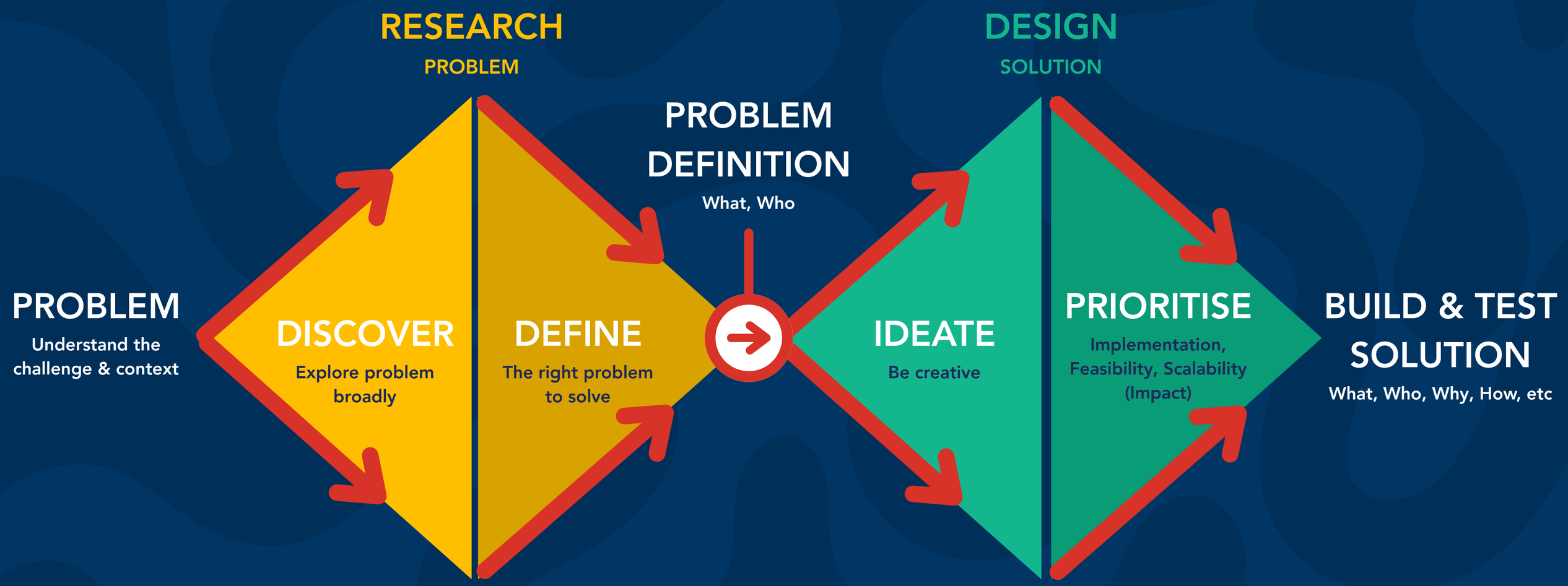
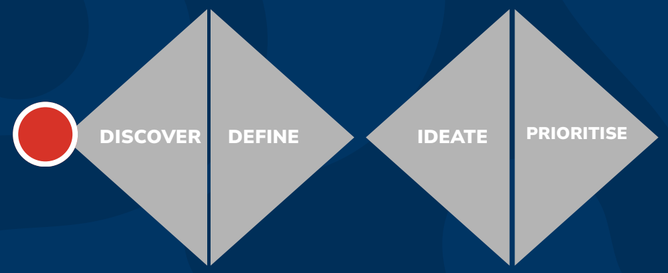
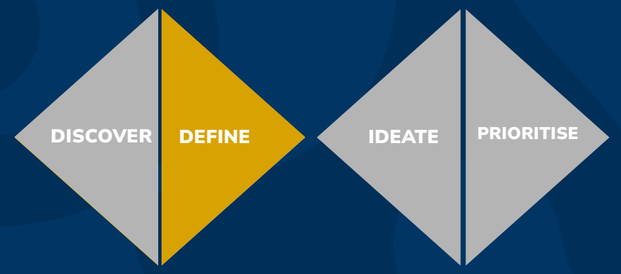
Innovation Process - Double Diamond





Problem

Hotels seem to waste a lot of food every day



Discover

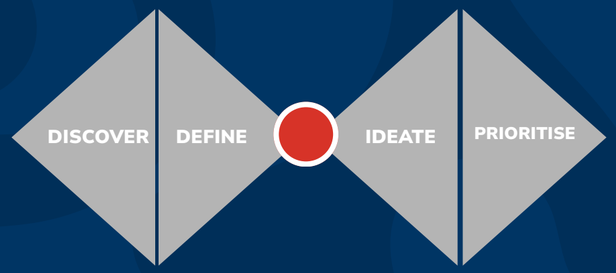
- Desk research
- Interviews
- Observation
- Understand users
- Identify needs
- Map the landscape

We spoke to restaurant managers from 3 hotels and learn a lot about their processes and the challenges they face

Define

- Synthesise insights
- Identify patterns
- Frame the probiem
- Meet target users
- Create opportunity areas

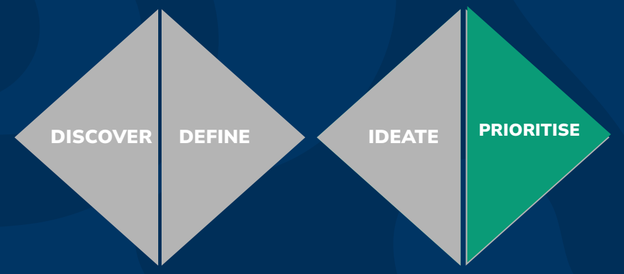
We shortlisted the top 3 most common challenges. Hotel buffets systematically over-prepare because the cost of running out is visible and immediate while the cost of throwing away is hidden, so tonnes of edible food are binned nightly.



Problem Defined

Hotel F&B operations need a practical way to reduce the food waste the restaurant produces from over-preparation.

How might we turn a hotel's unavoidable food waste into something of value?



Ideate

- Brainstorm ideas
- Explore solutions
- How might a working solution look like
- How might a prototype look like

We brainstormed 10 ways of reducing food waste - from reduction at source to composting what is inevitably created.

Prioritise

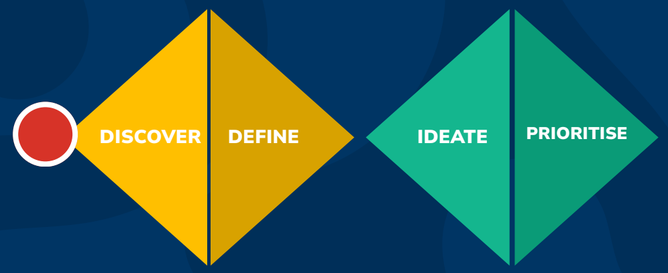
- Compare the different ideas
- Implementation: how, timeline, etc
- Feasibility: stakeholders involved, cost, etc
- Scalability: automated, manpower needed, budget, etc
- Potential impact: waste diverted

We landed on 2 possible prototypes:

1. Smart-bin to capture data on buffet leftovers
2. Composting network for restaurants and landscapers

For the solos: if you indicated interest to form teams, we'll put you into breakout rooms with other solos - introduce yourself + your interested waste problem

Your Turn!



10 min

Think & Research on your own

10 min

Discuss problems

Think, Research & Discuss

3 waste problems you have seen before

3 waste problems new to you

that you want to tackle

How big is the problem, who is involved, why is it currently like this

10 min

Discuss problem direction

What direction do you want to go in to:

- Conduct further research to make sure you fully understand the problem
- Brainstorm and prioritise ideas

Bonus: Industry Partner Challenge Statements

SUSS

Challenge Statement 1: Improving Food Waste Segregation: How might SUSS improve food waste segregation at selected campus outlets to reduce contamination and increase the amount of food waste that can be effectively composted?

Challenge Statement 2: Enhancing Compost Utilisation: How might SUSS develop sustainable and scalable ways to use or distribute the compost generated on campus, so that it benefits both the SUSS and wider community?

Challenge Statement 3: Reduction of food waste generated by customers: How might SUSS encourage the diners at the selected food outlets reduce food waste (e.g. order as per appetite, finish what's on plate).

Vidacity

"We run a few event spaces - our events generates food and packaging waste almost daily from deliveries, dining ware and excess food. Decision makers are often our partner and co-organisers. We'd like to explore ways to encourage or incentivize partners to adopt reusables, and participants to accept reusables or a bring-your-own culture at community events."

Kidztropic

"How might we design an engaging, play-based learning journey that helps preschool children aged 4 to 6 understand plastic waste and practise simple, sustainable habits in their everyday lives?

More details shared over email!

OPTIONAL Industry partner challenge statements for your consideration whether you want to tackle these problems to build your prototype. You can work on other problems too. If you want to work on any of these challenge statements, please let me know!

Singapore Deep-Tech Alliance

Next:
Discover & Define Problem

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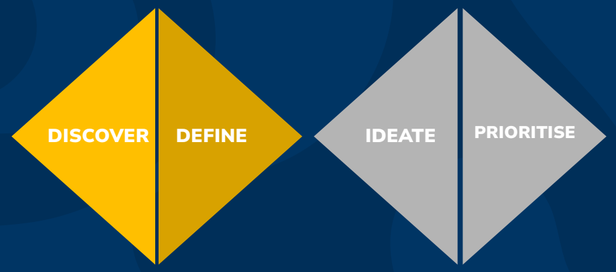
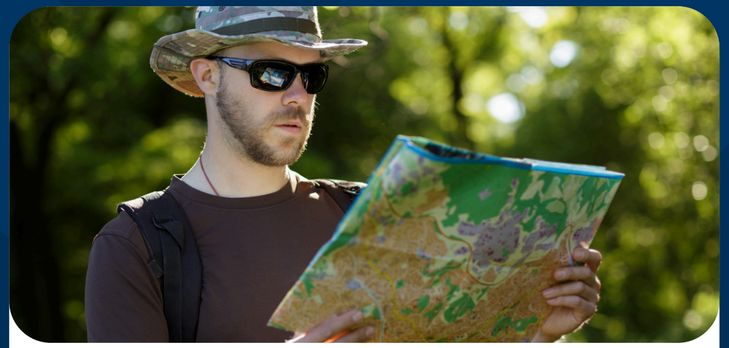
Discover & Define Problem





Explore

Mapping of system

Look

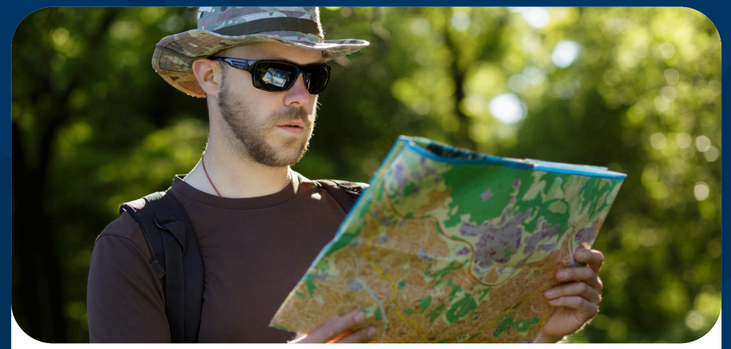
Observation

Ask

Understanding

- To avoid making the wrong assumptions
- Make sure you are tackling the right problem
- Have the foundation to build the right solution

Do you remember?
"Read the question carefully"



Explore

Mapping of system

Stakeholder Map

Everyone that is related to the waste

Solutions Scan

See what already exists and where the gaps are

Journey Map

Pinpoint the exact moment the waste is created

Look

Observation

AEIOU (Activities, Environments, Interactions, Objects, Users)

Waste data

Baseline: how much, where, how often

Ask

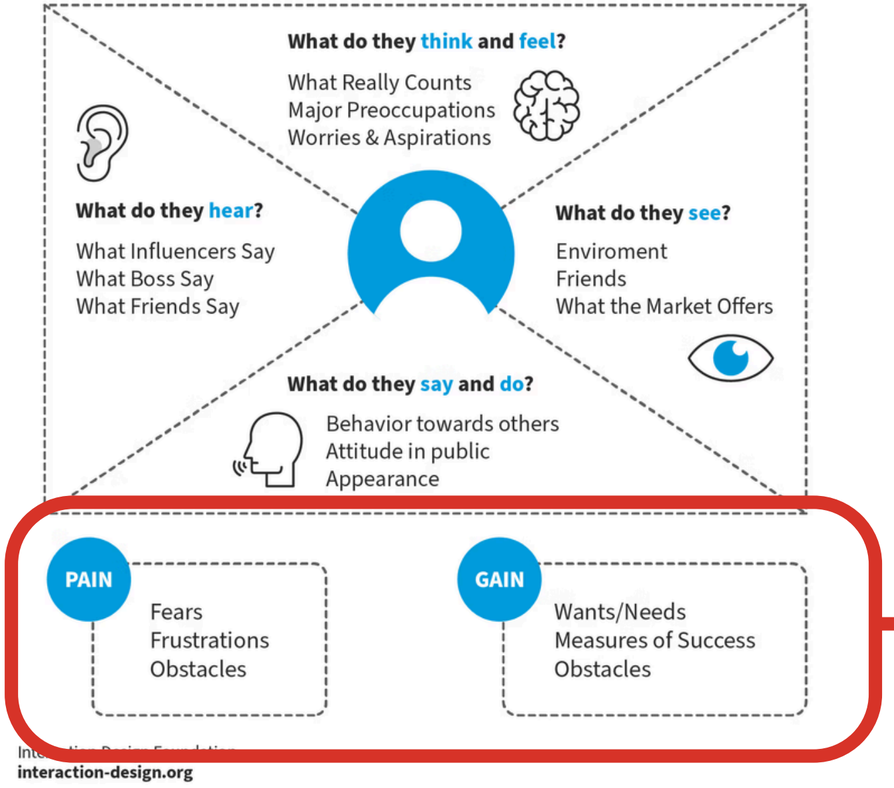
Understanding

User interviews

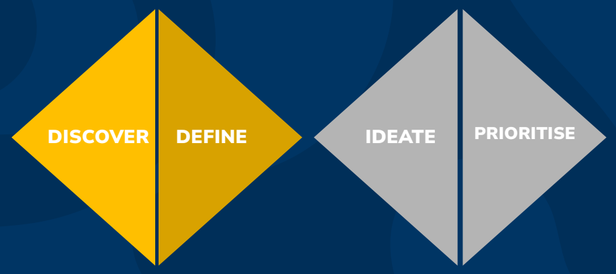
Questionnaire

Empathy map

Use this as a guide



Your solution should address these



Problem & Perspective

Problem You Want to Solve

- Environment
- Waste
- Carbon emissions
- Inefficiencies
- Future Generation

and more

Same?

No.

Problem of the User

- Manpower
- Money
- Managing customers
- Inefficiencies
- Convenience
- Regulations
- Integration
- Branding

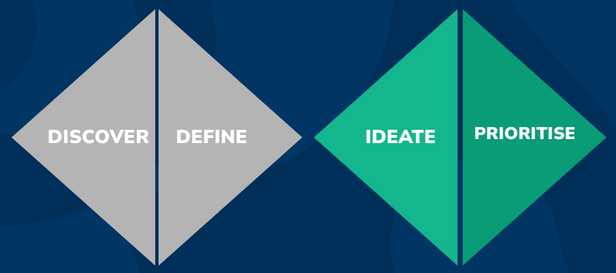
and more

Tips

- Sustainability is often **not** the top priority for people and businesses
- Tailor your "pitch" to the audience by highlighting the problems they care about
  - Reduce food waste vs reduce money wasted on ingredients that get thrown out
- Not every user has the same problem - more data, more confidence

How to conduct interviews - mistakes to avoid

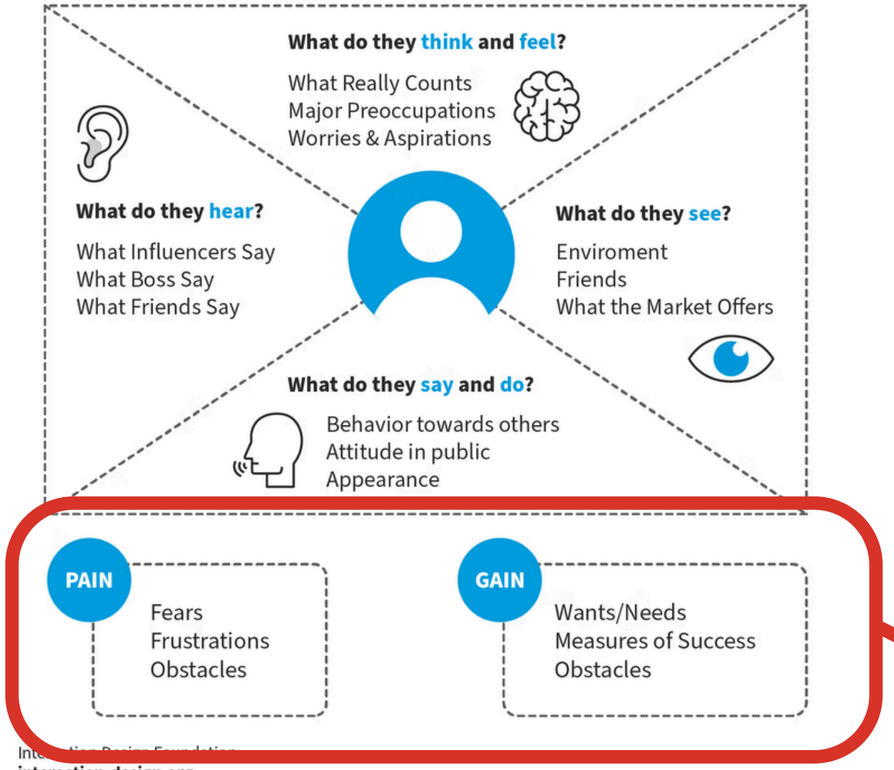
Ideas & Prioritising



Reminder

Empathy map

Use this as a guide



Tips

- Don't limit yourself from the start - bold and creative
- More impact if more people do a bit than few people do a lot
- Make it easy for the user - help them think of adoption frictions and mitigations
- How would you measure impact (waste diverted)

Your solution should address these

DELIVERABLES

Singapore Deep-Tech Alliance

Prototype

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Scale-up Plan

- Learnings, how you'd scale up / pivot your innovation if you could continue building this
- Take a longer term view to problem solving

Trash Diaries

- Log every kg diverted each week - weigh & photo
- Aim for 20 kg or more per team
- **Win Biggest Impact**!

People Log

- Log all individuals you engaged during your journey
- Share about SG Eco Loop, waste diversion journey, survey participants, test users, encourage recycling, etc

DELIVERABLE

Scale-Up Plan

- Slides submitted **by 1 Oct**, pdf format
- Optional - use selected slides as a supplementary asset for showcase

PROBLEM YOU TACKLED

- What problem are you solving, how big is it, who's the target user, who else is involved
- Why did you pick this problem
- What did you learn about the problem through this entire process?
- What do you still want to learn about this problem?

PROTOTYPE YOU BUILT & TESTED

- Describe your concept - what it would be like as a real-world solution, how it works
- Describe your prototype - what you actually managed to build, how it represents the concept
- How you tested it, with whom, where
- Findings and insights from the prototyping and testing process
- Impact - waste diverted using the prototype

REFLECTIONS

- What you learned through this journey - could be about sustainability, innovation, your specific problem, etc
- What was the biggest challenge and how did you handle it?
- Skills your team built, developed or improved

SCALING UP

- If you were to continue this project, what would you do - refine, pivot, etc
- Include:
  - Execution plan
  - Timeline for next 6 months
  - Stakeholders involved - how you would bring them in
  - Budget breakdown

Potential for committed teams to get post-programme support for further development & deployment

Tips

- **Photos** of the design, build and test process (prototype, your team, mentor, interviews, site visits, test users, etc) - add captions
- Illustrations
- Acknowledgements - mentors, industry partners

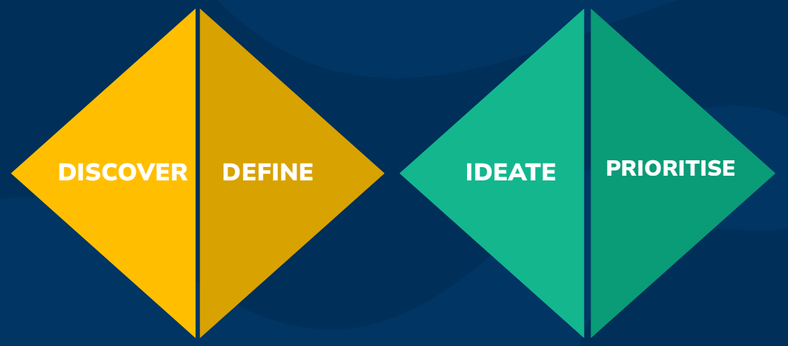
- **Use of AI:**
  - Bias against obvious AI output
  - Should not replace actual user interactions
  - You should be able to explain what is submitted

Wrapping Up Kickoff

Today:

Singapore's Waste Problems

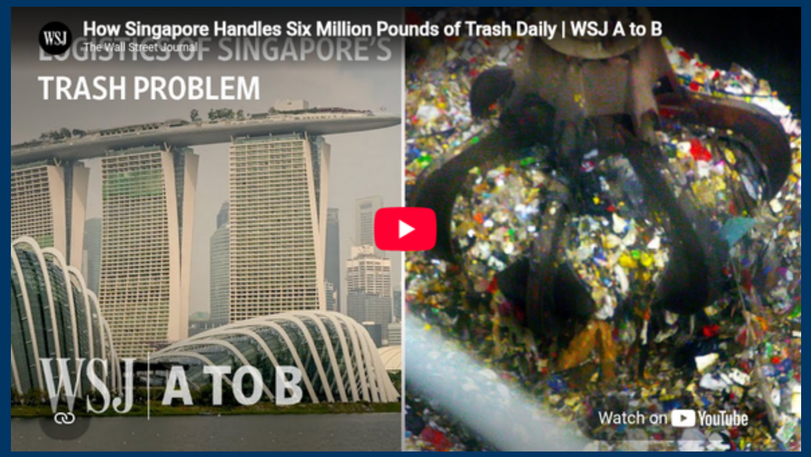
9 'R's



Come to Hackathon Day 1 (5 Sep) with these:

- **Clear Problem Statement**
  - What problem are you solving, how big is it, who's the target user, who else is involved
  - Why did you pick this problem
- **2-3 prioritised Ideas you want to further develop at the Hackathon**
  - Describe your concept - what it would be like as a real-world solution, how it works

Some recommended videos & articles to learn more about waste and recycling in Singapore



WSJ - **How Singapore Handles Six Million Pounds of Trash Daily**

CNA Insider - Do You Recycle? Why It Could Be Going to Waste

NEA - Singapore's Waste Generation Continues Its Downward Trend

Singapore Deep-Tech Alliance

See you at SG Eco Loop Hackathon Day 1!

Venue Change!

Lorong AI @ One-North
69 Ayer Rajah Cres., Level 3 Vidacity,
Singapore 139961

More questions?
Email: sgecoloop@sdta.org.sg
WhatsApp: +65 9019 5694
Jamie

5 September 2026, Saturday
9 am - 1 pm
Report: 8:45 am

Bring:

- Water bottle
- Pen & notebook / paper
- Laptop if have

For those of you joining solo and interested to form teams, please stay on

Forming Teams

- Any interest to continue conversations?
- WhatsApp group
- You can also discuss and decide later